Data, Decisions, and AI/ML Discoveries: Lab A

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Upload a single .py file to Canvas

1 Objectives

On completing this lab, you will be able to:

1. Write an object-oriented Python program.
2. Write output to the console.
3. Manipulate numeric data.
4. Utilize If/Else statements.
5. Utilize For loops.
6. Iterate through lists.

2 The Lab

Instructors submit final grades at the end of every term. Your task is to calculate the final grade of each student in a class using Python.

A student's letter grade is determined by the following scale, where x is their total score:

A $100 \geq x \geq 90$
B $90 > x \geq 80$
C $80 > x \geq 70$
D $70 > x \geq 60$
F $60 > x \geq 0$

Here are the final grades for 20 students in each of three categories:

Final Grades

Name	Homework	Quizzes	Exams
Alexis	80	89	92
Andrew	87	76	74
Ashley	98	84	88
Austin	92	93	87
Brandon	78	75	75
Chris	97	89	89
Eliza	100	91	72
Emily	100	95	95
Hannah	65	90	86
Jacob	71	76	85
Jessica	100	90	100
Joshua	91	91	91
Madison	13	38	38
Matthew	88	48	68
Michael	99	94	89
Nick	73	92	84
Sammy	94	84	71
Sarah	33	26	37
Taylor	93	89	90
Tyler	96	92	93

Suppose the categories are given weights.
Homework is weighted at 25%. Quizzes are weighted at 20%. Exams are weighted at 55%.
The total score for Alexis, for example, is given by $(80 * 0.25) + (89 * 0.20) + (92 * 0.55) = 88.4$.
"Rounding" in this class will always be done by taking the ceiling of the total score.
Hence, Alexis' final score for grading purposes is 89%. Her letter grade in the class is an B.

Suppose that, after a few years, weights are changed.
Homework is weighted at 60%. Quizzes are weighted at 20%. Exams are weighted at 20%.
The total score for Tyler, for example, is given by $(96 * 0.60) + (92 * 0.20) + (93 * 0.20) = 94.6$.
"Rounding" in this class will always be done by taking the ceiling of the total score.
Hence, Tyler's final score for grading purposes is 95%. His letter grade in the class is an A.

2.1 Deliverables

Determine the total score and final grade for each student in the table under both weighting schemes.
Your output should match my output. My output is located on the next page.

3 Ideal Output

```
Final Grades
Weights: Homework at 0.25; Quizzes at 0.2; Exams at 0.55;

Name        %       Grade
Alexis      89      B
Andrew      78      C
Ashley      90      A
Austin      90      A
Brandon     76      C
Chris       91      A
Eliza       83      B
Emily       97      A
Hannah      82      B
Jacob       80      B
Jessica     98      A
Joshua      91      A
Madison     32      F
Matthew     69      D
Michael     93      A
Nick        83      B
Sammy       80      B
Sarah       34      F
Taylor      91      A
Tyler       94      A

Final Grades
Weights: Homework at 0.6; Quizzes at 0.2; Exams at 0.2;

Name        %       Grade
Alexis      85      B
Andrew      83      B
Ashley      94      A
Austin      92      A
Brandon     77      C
Chris       94      A
Eliza       93      A
Emily       98      A
Hannah      75      C
Jacob       75      C
Jessica     98      A
Joshua      91      A
Madison     23      F
Matthew     76      C
Michael     96      A
Nick        79      C
Sammy       88      B
Sarah       33      F
Taylor      92      A
Tyler       95      A
```

4 Case Study

In Chile, students are not given a letter grade at the end of a course. Instead, students are given a number ranging from 1 to 7. 1 is the worst score, 7 is the best. To pass, students must earn a 4 or higher. Suppose that a student's grade is determined by the scale below, where x is their total score:

Seven: $100 \geq x \geq 85$

Six: $85 > x \geq 70$

Five: $70 > x \geq 55$

Four: $55 > x \geq 40$

Three: $40 > x \geq 30$

Two: $30 > x \geq 15$

One: $15 > x \geq 0$

The code provided in the file `labA_case_study.py` addresses this situation on the same set of students. Take a look at it if you're feeling stuck.